

More Modules!

Exercise 1.

- (a) Suppose M is a free module. Show that M is projective.
- (b) Prove that if P is projective, then any exact sequence $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ splits on the right.
- (c) Any module is a quotient of a free module. A universal, but very wasteful, way to accomplish this would be to make F have one basis element e_m for every $m \in M$, and take the homomorphism $F \rightarrow M$ that sends e_m to m . Prove that if P is a module with the property that any exact sequence $0 \rightarrow A \rightarrow B \rightarrow P \rightarrow 0$ splits, then P is a direct summand of a free module. That is, there is some module S so that $P \oplus S$ is free.
- (d) Show that if P is projective and $P = S \oplus T$, then S is also projective.
- (e) Combine parts (a) and (d) to conclude that if P is a direct summand of a free module, then P is projective.

Solution 1.

- (a) Since M is free, say $M = R^{\oplus n}$ with basis elements e_i (everything here also works for an infinite rank free module, but it's more annoying to write down). Suppose $f: B \rightarrow C$ is surjective, and $\phi: M \rightarrow C$ is any map. Define a map $\tilde{\phi}: M \rightarrow B$ as follows: for each basis element e_i , choose $b_i \in f^{-1}(\phi(e_i))$, and define $\tilde{\phi}(e_i) = b_i$. This gives a homomorphism which is evidently a lifting of ϕ .
- (b) Given such an exact sequence, we can choose $\text{id}: P \rightarrow P$. Since P is projective, there is some $s: P \rightarrow B$ which lifts id , and this s is exactly a section of the map $B \rightarrow P$.
- (c) We know there is some free module F with a surjection $\pi: F \rightarrow P$. From the previous part, we know that if P is projective then we can find a section $s: P \rightarrow F$, so that $F \cong P \oplus \ker \pi$. Thus, we may choose $S = \ker \pi$.
- (d) Suppose $f: B \rightarrow C$ is surjective, and suppose $\phi: S \rightarrow C$ is any map. Since $P = S \oplus T$, we may extend ϕ to a map $\phi': P \rightarrow C$ by defining $\phi'(s, t) = \phi(s)$. Now, since P is projective, we can find a lift of ϕ' , $\tilde{\phi}': P \rightarrow B$. Let $\tilde{\phi}$ be the restriction of $\tilde{\phi}'$ to S , we claim that $\tilde{\phi}$ is a lift of ϕ . Indeed, $f(\tilde{\phi}(s)) = f(\tilde{\phi}'(s, 0)) = \phi'(s, 0) = \phi(s)$. Thus, S is projective.
- (e) This follows immediately.

Exercise 2.

- (a) Let $R = \mathbb{Z}/6\mathbb{Z}$, and let $M = \mathbb{Z}/3\mathbb{Z}$. Then M is an R -module under the action $r \cdot m = rm$ for $r \in R, m \in M$. Prove that M is projective, but not free.
- (b) Let $R = M_2(\mathbb{R})$, and let $M = \mathbb{R}^2$. Then M is a left R -module under the action $A \cdot v = Av$ for $A \in R$ and $v \in M$. Prove that M is a projective left R -module, but not a free left R -module.

- (c) Let R be a PID. What are all the finitely-generated projective R -modules?
- (d) Let M be a projective module over a commutative ring R , and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

Solution 2.

- (a) We have that M is a direct summand of the free module R , since $R \cong \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. However, any free module has order a power of 6, while $\mathbb{Z}/3\mathbb{Z}$ has order 3, so $\mathbb{Z}/3\mathbb{Z}$ cannot be free.
- (b) Notice that $M_2(\mathbb{R}) \cong V \oplus V$ as a left module over itself, where the first V is the first column of the matrix and the second V is the second column. Thus, V is a direct summand of a free module, therefore is projective. On the other hand, any free module has dimension a multiple of 4, but V has dimension 2, so V is not free.
- (c) A finitely-generated module over a PID is projective iff it is free, because torsion modules are never summands of free modules. (Alternatively, any submodule of a free module over a PID is itself free. This is actually true even for infinitely generated free modules, but it's obnoxious to prove.)
- (d) Since M is projective, by part (a), we have that $M \oplus N \cong R^d$ for some R -module N and some $d \geq 1$. Therefore, we have a split SES $0 \rightarrow N \rightarrow R^d \rightarrow M \rightarrow 0$. Since localization is exact, we get that $0 \rightarrow S^{-1}N \rightarrow S^{-1}(R^d) \rightarrow S^{-1}M \rightarrow 0$ is also a split SES. Thus, $S^{-1}M \oplus S^{-1}N \cong S^{-1}(R^d) \cong R[S^{-1}]^d$, so $S^{-1}M$ is a projective $R[S^{-1}]$ -module by part (a).

Exercise 3.

- (a) Prove that a free module is flat. (With a little more work, one can use this to show that a projective module is flat.)
- (b) Let R be an integral domain, and let M be a flat R -module. Prove that M has no torsion, i.e. there are no $m \in M \setminus \{0\}$ and $r \in R \setminus \{0\}$ so that $rm = 0$. (Hint: there is an exact sequence $0 \rightarrow R \xrightarrow{\times r} R \rightarrow R/(r) \rightarrow 0$.)
- (c) Let R be a PID. What are all the finitely-generated flat R -modules.
- (d) Let $S \subset R$ be a multiplicative set. Show that $S^{-1}R$ is a flat R -module. (Hint: $S^{-1}R \otimes_R M = S^{-1}M$.)

Solution 3.

- (a) For any module A , we have that $A \otimes_R R^{\oplus n} \cong A^{\oplus n}$. Given any exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, tensoring with $R^{\oplus n}$ gives $0 \rightarrow A^{\oplus n} \rightarrow B^{\oplus n} \rightarrow C^{\oplus n} \rightarrow 0$, which is still exact.
- (b) Tensoring the exact sequence in the hint with M gives an exact sequence $0 \rightarrow M \xrightarrow{\times r} M \rightarrow M/rM \rightarrow 0$. Since this sequence is exact, we know that multiplication by $r \neq 0$ is injective, and in particular is not the zero map on M for any r . Thus, M has no torsion.
- (c) Once again, a f.g. flat module over a PID turns out to be free. This time, however, it is not true for infinitely generated modules, as we'll see in the next problem.

(d) Since $S^{-1}R \otimes_R M = S^{-1}M$, and since localization is itself exact, we have that $S^{-1}R$ is flat.

Exercise 4. Part (d) of the previous question implies that $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module. Let $A = \mathbb{Z}^{\oplus \mathbb{N}}$ be a countably-generated free $\mathbb{Z}$ -module with basis $\{e_1, e_2, \dots\}$, and consider the homomorphism $f: A \rightarrow \mathbb{Q}$ which sends $e_i \mapsto \frac{1}{i}$. Prove that f is surjective, but there is no section $s: \mathbb{Q} \rightarrow A$. Hence, $\mathbb{Q}$ is not projective (nor indeed free).

Solution 4. The map f is obviously surjective. Suppose f had a section s . For any fraction $\frac{a}{b}$, we have that $b \cdot \frac{a}{b} - a \cdot 1 = 0$, so $bs(\frac{a}{b}) - as(1) = 0$, so $s(\frac{a}{b})$ and $s(1)$ are linearly dependent. Thus, $\text{im } s$ has rank at most 1. But if $\text{im } s$ had rank 1, then it would give a map $\mathbb{Q} \rightarrow \mathbb{Z}$, and there are no such homomorphisms because no element of $\mathbb{Z}$ is divisible by every integer. Thus, s would have to be the 0 map, but that isn't a section of f .

Exercise 5. A (not necessarily commutative) ring R is **simple** if the only two-sided ideals of R are (0) and all of R .

A (not necessarily commutative) ring R is a **division ring** if for every nonzero $r \in R$ has a multiplicative inverse r^{-1} .

- (a) Prove that a division ring is simple.
- (b) Prove that a nonzero commutative simple ring is a field.
- (c) Let $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ be Hamilton's quaternions, which has multiplication defined by the relationships $i^2 = j^2 = k^2 = ijk = -1$ and $ij = -ji$. Show that $\mathbb{H}$ is a division ring.
- (d) Show that $M_2(\mathbb{R})$ is a simple ring but not a division ring.

Solution 5.

- (a) If D is a division ring, and $I \subset D$ is a nonzero ideal, then taking any nonzero $i \in I$, we have that $1 = i^{-1}i \in I$, hence $I = D$. (Indeed, this shows moreover that a division ring doesn't have any 1-sided ideals aside from (0) and (1) .)
- (b) If a commutative ring has only the ideals (0) and (1) , then for any $r \neq 0$, $(r) = (1)$, and so there is some r^{-1} so that $r^{-1}r = 1$, hence r is invertible. Thus, the ring is actually a field.
- (c) For any $a + bi + cj + dk \in \mathbb{H}$, $(a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2 \in \mathbb{R}$. Thus, the element $\frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}$ is an inverse to $a + bi + cj + dk$.
- (d) Suppose I is a nonzero two-sided ideal of $M_2(\mathbb{R})$, and take any matrix $A \in I$. Write out the entries of A :

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where not all of a, b, c, d are 0. Without loss of generality, suppose $a \neq 0$. Then, notice that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a^{-1} & 0 \\ 0 & 0 \end{pmatrix} \in I.$$

Furthermore,

$$\begin{aligned} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in I \\ \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \in I \\ \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in I. \end{aligned}$$

Thus, any linear combination of these four matrices is in I , but these four matrices span $M_2(\mathbb{R})$, so $I = M_2(\mathbb{R})$.

On the other hand, $M_2(\mathbb{R})$ isn't a division ring because there are nonzero matrices which aren't invertible. We just wrote down 4 of them, in fact.

Exercise 6. Similar to the above, a left (right) module M over a ring R is **simple** if the only left (right) submodules of M are 0 and all of M . (A non-simple ring can have simple modules.)

A module M is **cyclic** if there is some $m \in M$ so that the module generated by m is all of M .

- Prove that a simple module is cyclic. (Hint: choose *any* nonzero $m \in M$.)
- Let R be a ring, and let $\mathfrak{m}$ be a maximal left ideal of R . Show that $R/\mathfrak{m}$ is a simple left R -module.
- Now let M be any simple left R -module. By part (a), $M = Rm$ for some $m \in M$. Show that $\text{Ann}(m)$ is a maximal left ideal of R . Conclude that every simple left module over R is of the form $R/\mathfrak{m}$. (Hints: (i) Suppose $I \supsetneq \text{Ann}(m)$, and take $i \in I \setminus \text{Ann}(m)$. What does part (a) say about im ? (ii) If $rm = m$, then $r - 1 \in \text{Ann}(m)$.)

Solution 6.

- Let $m \in M$ be nonzero. Then the submodule generated by m is a nonzero submodule of M , but the only nonzero submodule is all of M , so m generates all of M .
- A left R -submodule of $R/\mathfrak{m}$ is the same thing as a left ideal of $R/\mathfrak{m}$, which is in bijection with ideals of R which contain $\mathfrak{m}$. However, since $\mathfrak{m}$ is maximal, the only such ideals are $\mathfrak{m}$ and (1) , which correspond to the submodules 0 and $R/\mathfrak{m}$. Thus, $R/\mathfrak{m}$ is a simple module.
- Suppose $I \supsetneq A$, and choose $i \in I \setminus A$. Then im is a nonzero element of M , and hence also generates all of M . Thus, there is some $r \in R$ so that $r(im) = m$. But then $ri - 1 \in \text{Ann}(m) \subset I$, so $1 = ri - (ri - 1)$ is a difference of two elements of I , therefore is in I .

Exercise 7. (August 2020 Problem 1) On this problem, no justification is required; only the answer will be graded.

- Give an example of a non-diagonalizable matrix in $\text{GL}_3(\mathbb{F}_p)$ (for some p).
- Give an example of a nontrivial proper normal subgroup of $\text{GL}_3(\mathbb{F}_p)$ (for some p).

- (c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.
- (d) Give an example of a $\mathbb{Z}$ -module which is flat but not free.
- (e) Give an example of an exact sequence of abelian groups which is not split.

Solution 7.

- (a) Let $p = 2$ and consider $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.
- (b) Let $p = 3$. Then $\mathrm{SL}_3(\mathbb{F}_3)$ is a proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_3)$.
- (c) Let I be the ideal of matrices of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$. Then I is a left ideal but not a right ideal.
- (d) $\mathbb{Q}$
- (e) $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$

Exercise 8. (January 2024 Problem 4, notation altered) Let R be a commutative ring, and let A and B be commutative R -algebras.

- (a) Prove that if B is flat over R , and f is a non zero divisor in A , then $f \otimes 1$ is a non zero divisor in $A \otimes_R B$.
- (b) Prove that the flatness condition above is necessary by giving an example where part (a) fails if B is not flat over R .

Solution 8.

- (a) We need to figure out a way to use the flatness of B . Flatness wins us only one thing: it tells us that if $M \hookrightarrow N$ is an injection of R -modules, then $M \otimes_R B \rightarrow N \otimes_R B$ is still an injection. This suggests we should try to phrase the condition “ f is a non zero divisor” in terms of some map being injective. This we can do: f being a non zero divisor is equivalent to saying that the “multiplication by f ” map $A \xrightarrow{f} A$ is injective. (Check for yourself that this is in fact a homomorphism of R -modules.) Having noticed this rephrasing we are now done: since B is flat, the map $A \otimes_R B \xrightarrow{f \otimes 1} A \otimes_R B$ is injective, which is the same as saying $f \otimes 1$ is a non zero divisor.
- (b) A good example of a non-flat module is a module that has torsion. So, for example, we could take $R = A = \mathbb{Z}$, and say $B = \mathbb{Z}/2\mathbb{Z}$, and the question becomes “is there a non zero divisor in $\mathbb{Z}$ that becomes a zero divisor in $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$ ”. And of course there is. For example, 2 is a non zero divisor in $\mathbb{Z}$ which becomes 0 in $\mathbb{Z}/2\mathbb{Z}$. (If you use the definition by which a zero divisor must be different from 0: first of all, cut it out, but also you could instead take say $B = \mathbb{Z}/4\mathbb{Z}$.)

Exercise 9. (January 2021 Problem 3) (This was on WS 3.)

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

Solution 9.

- (a) Let M be a projective R -module, where R is an integral domain. Then M is a direct summand of a free R -module, i.e. $M \oplus N = R^d$ for some R -module N and $d \in \mathbb{N}$. Let $m \in M$ be a nonzero element, and suppose that $am = 0$ for some $a \in R$. We will show that $a = 0$. We can write $m = (r_1, \dots, r_d)$ with $r_i \in R$. Then $am = (ar_1, \dots, ar_d) = 0$ implies that $ar_i = 0$ in R for all i . Since R is a domain, this means that either $a = 0$ or $r_i = 0$ for all i . Since $m \neq 0$, we must have that $a = 0$. Thus, M is torsion free.
- (b) (i) This is not true. For example, consider the SES of $\mathbb{Z}$ -modules $0 \rightarrow \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$, where the left map is multiplication by 2 and the right map is the natural quotient map. (Or similarly, $0 \rightarrow k[x] \xrightarrow{\cdot x} k[x] \rightarrow k \rightarrow 0$.) Then $\mathbb{Z}$ is projective, but $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$ -module. One way to see this directly is that if $\pi : \mathbb{Z}/4\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the map given by modding out by 2 and $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is the identity map, then there is no map that makes the following diagram commute:

$$\begin{array}{ccc} & & \mathbb{Z}/2\mathbb{Z} \\ & \swarrow & \downarrow \\ \mathbb{Z}/4\mathbb{Z} & \longrightarrow & \mathbb{Z}/2\mathbb{Z} \end{array}$$

This because the only possible maps are the zero map and multiplication by 2, and neither of these gives the identity map when composed with π . Another way to see that $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$ -module is to use that every projective module is flat and show that it is not flat.

- (ii) This is true. If M_3 is projective, then the SES is split exact, so $M_2 \cong M_1 \oplus M_3$. Recall that being projective is equivalent to being a direct summand of a free module. If M_2 is projective, then M_1 is as well since it is a direct summand of M_2 which is a direct summand of a free module.

Exercise 10. (January 2021 Problem 2) Let k be a field, and let $R = M_2(k)$, the non-commutative ring of 2×2 matrices over k .

- (a) Give examples of a simple left R -module M and a simple right R -module N .
- (b) Can you find another simple left R -module M' that is not isomorphic to M ? (Either find one or explain why there is none such.)

- (c) Compute $\dim_k(N \otimes_R M)$.

Solution 10.

- (a) Luckily, the ring R has a module that we already know and love: the vector space k^{2n} . Indeed, let v be a nonzero vector, and $v' \in k^n$ an arbitrary vector, then there exists a linear transformation that takes v to v' (and say, takes some complementary subspace to $\langle v \rangle$ to 0). Thus, $Rv = k^2$, and k^2 is a simple left module.

For a simple right module, we could take k^2 viewed as row vectors rather than column vectors with matrix multiplication on the right, or equivalently we could take k^2 viewed as column vectors with $v \cdot A = A^T v$, or equivalently we could take the dual vector space $\text{Hom}_k(k^2, k)$ with $\phi \cdot A = \phi \circ A$. Confirm to yourself that these are the same.

- (b) (Previously, there was a subtle error in my argument here. Thanks to Gabriela Brown for bringing it to my attention.) If I were guessing the answer this problem based only on the phrasing, I would guess that the answer is “none exists”. Let us show that is the case.

Suppose we have a simple left R -module M' . We will show that M' is isomorphic to k^n . Since M' is simple, it is generated by any nonzero $m' \in M'$. So, we can define an R -module homomorphism $\phi: M' \rightarrow M$ by specifying where to send just m' and then extending R -linearly. The only thing we need to do to make sure the resulting map is a homomorphism is to make sure that whenever $Am' = 0$, we also have $A\phi(m') = 0$. So, let $I = \{A \in R : Am' = 0\}$ be the annihilator of m' , which is a left ideal of R . Any matrix in I must have nontrivial kernel, because an invertible matrix cannot kill a nonzero element. Indeed, the matrices in I must all have a nontrivial mutual kernel, because if they didn't then a linear combination of them would have trivial kernel, which we just said cannot happen. Thus, there is a nonzero vector $v \in k^n$ that is in the kernel of every matrix in I . We define our map ϕ by sending $m' \mapsto v$.

The image of this map is a nontrivial submodule of k^n , hence is all of k^n since k^n is simple, and so this map is onto. Notice now that its kernel is a submodule of M' , hence is either 0 or all of M' . Since this map is not the zero map, the kernel cannot be all of M' , so is 0, and so this is an isomorphism between M' and k^n . (There's actually some cool afterlife to this fact: look up “Morita equivalence”.)

- (c) (Previously I had an incorrect solution here. Thanks to Abbi Jones, Hannah Ashbach, and Bella Finkel for pointing out my error.) Let's view M as k^2 viewed as column vectors and matrix multiplication on the left, and N as k^2 viewed as row vectors and matrix multiplication on the right. Let e_1 be the first standard basis (column) vector for M . Then, given any other vector $v \in M$, we can choose the matrix A which has v as its first column and all other entries 0, and this matrix satisfies $Ae_i = v$. We can therefore rewrite any simple tensor $u \otimes v = u \otimes Ae_i = uA \otimes e_1$. Now, since $u \in k^2$, we should view u as a row vector, and by our choice of A , $uA = (\lambda, 0, \dots, 0)$ for some $\lambda \in k$. Thus, every vector in $N \otimes_R M$ is equivalent to a vector of the form $(\lambda, 0, \dots, 0) \otimes e_1$, and projection onto the first factor gives an isomorphism between $N \otimes_R M$ and $\langle (\lambda, 0, \dots, 0) \rangle$, so $\dim_k N \otimes_R M = 1$.

Here's another, more abstract way of putting it. Any simple left (right) module is isomorphic to $R/\mathfrak{m}$ for some left (right) maximal ideal $\mathfrak{m}$. I know of some maximal ideals of $M_n(\mathbb{C})$: there's

the maximal left ideal I consisting of matrices where the first column contains all 0s, and the maximal right ideal J consisting of matrices where the first row contains all 0s. So, I can write $M = R/I$, $N = R/J$. Then, we have a theorem that tells us that for any right module N and left ideal I , $N \otimes_R R/I \cong N/NI$. Thus, we get that $N \otimes_R M \cong N/NI = R/(I + J)$ where by $I + J$ I mean the additive subgroup of R consisting of matrices that can be written as a sum of matrices from I and matrices from J . The subgroup $I + J$ is just the vector subspace of R consisting of matrices with a 0 in the $(1, 1)$ position, and when we mod out by this subgroup we get a 1-dimensional vector space (parametrized by the possible values in the $(1, 1)$ position of a matrix).